Complex Numbers: Modulus-Argument Form

Questions 1–54

Question 1. Given that $z = 4 - 2i$ and $w = -1 + 3i$, find

a) $|z|$
b) $|w|$
c) $|2w|$
d) $\left|\frac{z}{w}\right|$

Question 2. Find the exact value of each of the following:

a) $\arg(1 + i)$
b) $\arg(1 + \sqrt{3}i)$
c) $\arg(-2i)$
d) $\arg(-2 - 2i)$
e) $\arg(-3)$

Question 3. Write each of the following in the form $a + bi$ where a and b are real numbers:

a) $3\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$
b) $\frac{1}{4}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$
c) $2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$
d) $7\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$
e) $12\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right)$
f) $\frac{\sqrt{2}}{4}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$

Question 4. Write each of the following complex numbers into the form $a + bi$ where a and b are real numbers, giving your answers in exact form:

a) $|z| = 2$ and $\arg z = \frac{\pi}{3}$
b) $|z| = 1$ and $\arg z = \frac{\pi}{4}$
c) $|z| = 4$ and $\arg z = -\frac{2\pi}{3}$

Question 5. Write each of the following complex numbers in modulus-argument form. Give the argument in radians in exact form.

a) $3 + 3i$
b) $2i$
c) -4
d) $-1 + \sqrt{3}i$
e) $2 - 2i$
f) $-\sqrt{3} - i$
g) $-7 + 7i$
h) $3 - \sqrt{3}i$

Question 6. Write each of the following complex numbers in modulus-argument form, giving the argument to 3 significant figures.

a) $1 + 5i$
b) $-2 + 5i$
c) $4 - i$
d) $-3 - 7i$
e) $\sqrt{2} + 6i$
f) $-1 + \frac{3}{4}i$
g) $2 - \sqrt{3}i$
h) $-1 - \frac{1}{2}i$

Question 7. Ava is finding the principal argument form of $-\sqrt{3}-i$. She writes the following:

$$r=\sqrt{(-\sqrt{3})^2+(-1)^2}=\sqrt{3+1}=2$$

$$\tan\theta=\frac{-1}{-\sqrt{3}}=\frac{1}{\sqrt{3}}$$

$$\theta=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)=\frac{\pi}{6}$$

So $-\sqrt{3}-i=2\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)$

a) Identify and explain the mistake in Ava's working.

b) Find the correct modulus-argument form of $-\sqrt{3}-i$

Question 8. Write each of the following into principal argument form:

a) $4\left(\cos\frac{5\pi}{2}+i\sin\frac{5\pi}{2}\right)$

b) $2\left(\cos\frac{\pi}{3}-i\sin\frac{\pi}{3}\right)$

c) $3\left(\cos\frac{10\pi}{3}-i\sin\frac{10\pi}{3}\right)$

d) $-5\left(\cos\frac{10\pi}{3}+i\sin\frac{10\pi}{3}\right)$

Prerequisite knowledge for Q9-Q11: Complex Conjugates

Question 9. Express each of the following complex numbers in modulus-argument form, giving the argument correct to 3 significant figures:

a) $\frac{1}{1-3i}$
b) $\frac{1}{1+3i}$
c) $\frac{5}{2-3i}$
d) $\frac{-7+i}{7-2i}$

Question 10. State the complex conjugate of

$$z = 3\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$$

Question 11. It is given that $z = 4 - 5i$

a) Write z in modulus-argument form, giving the argument to 3 significant figures.

b) Hence state z^* in modulus-argument form.

Question 12. Determine the solutions to each of the following equations, giving your answer in modulus-argument form:

a) $z^2 - 2z + 2 = 0$

b) $9z^2 - 3z + 1 = 0$

c) $z^2 + 8z + 64 = 0$

d) $k^2z^2 - 2kz + 2 = 0$
where k is a positive real number.

Question 13. Given that $z_1 = 2 + 3i$ and $z_2 = iz_1$

a) write z_1 and z_2 in modulus-argument form giving your answer to 3 significant figures.

b) Plot z_1 and z_2 on the same diagram.

c) Describe the geometrical transformation that maps z_1 onto z_2

Question 14. Given that $z_1 = 1 + 2i$ and $z_2 = 2iz_1$

a) write z_1 and z_2 in modulus-argument form giving your answer to 3 significant figures.

b) Plot z_1 and z_2 on the same diagram.

c) Describe the geometrical transformation that maps z_1 onto z_2

Question 15. Given that $z_1 = 3 + 2i$ and $z_2 = (1 + i)z_1$

a) write z_1 and z_2 in modulus-argument form giving your answer to 3 significant figures.

b) Plot z_1 and z_2 on the same diagram.

c) Describe the geometrical transformation that maps z_1 onto z_2

Question 16.

a) Given $z_1 = 3 + 2i$ and $z_2 = 1 - 5i$, verify that $|z_1 + z_2| \leq |z_1| + |z_2|$

b) Explain geometrically why for two complex numbers z and w it is always true that

$$|z + w| \leq |z| + |w|$$

Question 17. It is known that

$$\arg(a+bi) = \frac{\pi}{4}$$

for real numbers a and b, find

a) $\arg(a-bi)$
b) $\arg(-a-bi)$
c) $\arg(b+ai)$
d) $\arg(b-ai)$

Question 18. Given that

$$\arg(a+bi) = -\frac{\pi}{3}$$

for real numbers a and b, find

a) $\arg(a-bi)$
b) $\arg(-a-bi)$
c) $\arg(b+ai)$
d) $\arg(b-ai)$

Question 19. Two complex numbers z and w on an Argand diagram are labelled A and B respectively. In each case, find the angle AOB giving your answer to 3 significant figures.

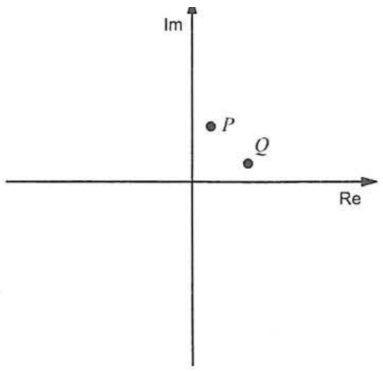
a) $z = 4 + 3i$, $w = 1 + 2i$
b) $z = 2 + 2i$, $w = -3 + 6i$
c) $z = -4 + 3i$, $w = 1 - i$

Question 20. Find the area of the triangle formed by vertices 0, z_1 and z_2 in each of these cases:

a) $z_1 = 2 + i$ and $z_2 = 1 + 3i$
b) $z_1 = 1 + 2i$ and $z_2 = -2 + 4i$
c) $z_1 = -5 + 3i$ and $z_2 = 2 - 3i$

Question 21. Given that $z_1 = 4 - i$, $z_2 = 6 - 8i$ and $z_3 = -4i$, find the exact area of the quadrilateral formed by the vertices of 0, z_1, z_2 and z_3

Question 22. Two complex numbers z_1 and z_2 are labelled as P and Q respectively.



a) On a copy of the Argand diagram, add a third point $z_1 + z_2$ and label the point R
b) Given that $OPRQ$ forms a rhombus, prove that $\arg(z_1 + z_2) = \frac{1}{2}(\arg z_1 + \arg z_2)$

Multiplying & Dividing in Modulus-Argument Form

Question 23. It is known that

$$z = 3\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

and

$$w = 5\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

Write the following complex numbers in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) zw
b) $\frac{z}{w}$
c) $\frac{w}{z}$
d) z^2
e) w^3

Question 24. It is known that

$$z = 2(\cos 1.4 + i\sin 1.4)$$

and

$$w = 4(\cos 0.7 + i\sin 0.7)$$

Write the following complex numbers in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) zw
b) $\frac{z}{w}$
c) $\frac{w}{z}$
d) w^2
e) w^2z
f) $\frac{z^2}{w^2}$

Question 25. It is known that

$$z = 7\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

and

$$w = 10\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right)$$

Write the following complex numbers in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) zw
b) $\frac{z}{w}$
c) $\frac{w}{z}$
d) $2w^2$

Question 26. For $z = 1 + i$ and $w = \sqrt{3} + i$, find zw^2 in principal argument form.

Question 27. Given that

$$z = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

and

$$w = 5\left(\cos\frac{4\pi}{5} + i\sin\frac{4\pi}{5}\right)$$

write the following complex numbers in principal modulus-argument form:

a) zw
b) $\frac{z}{w}$
c) w^3
d) $(1+i)z$

Question 28. The complex numbers z_1 and z_2 are given by

$$z = 2\left(\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right)$$

and

$$w = \frac{1}{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$$

Write the following complex numbers in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) zw
b) $\frac{z}{w}$
c) $(zw)^2$
d) $(1+\sqrt{3}i)w$

Question 29. It is known that

$$z = \sqrt{3} + \sqrt{3}i \text{ and } w = 4i$$

Write the following complex numbers in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) z
b) w
c) Hence, find zw^2 in principal modulus-argument form.

Prerequisite knowledge for Q30: Complex Conjugates

Question 30. For the complex number z, it is known that $|z| = 3$ and $\arg z = \frac{2\pi}{3}$, find each of the following in **both** modulus-argument form and in the form $a + bi$ where a and b are real numbers:

a) z
b) z^*
c) z^2
d) $\frac{z}{z^*}$

Question 31. For the complex number z, in each case state the argument of z^2

a) $\arg(z) = \frac{\pi}{4}$

b) $\arg(z) = -\frac{\pi}{3}$

Question 32. In each case, express $\frac{z}{w}$ in the form $a + bi$ where a and b are real numbers given to 3 significant figures:

a) $|z| = 2$, $\arg z = \frac{\pi}{3}$ and $|w| = 3$, $\arg w = \frac{\pi}{4}$

b) $z = 1 + i$, $|w| = 2$ and $\arg w = \frac{\pi}{6}$

Question 33. The complex number z is given by $5 - 2i$ and another complex number w exists such that

$$wz = \sqrt{290}(\cos 0.8 + i \sin 0.8)$$

a) Find $|w|$
b) Find $\arg w$ giving your answer to 3 significant figures.

Question 34. Given that

$$z_1 = -\frac{\sqrt{3}}{2} - \frac{3}{2}i$$

$$|z_1 z_2| = \sqrt{30}$$

and

$$\arg(z_1 z_2) = \frac{\pi}{4}$$

a) Write z_1 in the form $r(\cos\theta + i\sin\theta)$
b) Find z_2 in the form $a + bi$ where a and b are real numbers.

Question 35. Two complex numbers, z and w may be represented on an Argand diagram where

$$\text{Im}(z) > \text{Im}(w)$$

The angle between them is $\frac{\pi}{3}$ radians and it is known that $|3z| = |w|$
Find $\frac{w}{z}$ in the form $a + bi$ where a and b are exact real numbers.

Question 36. The complex number z is

$$-k + ki$$

where k is a positive integer.

a) Express z is modulus-argument form.

b) Given that

$$w = 2\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

and

$$\text{Re}(zw) = -20$$

find the value of k

Question 37. Given that

$$z = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

find iz^3 writing your answer in principal modulus-argument form.

Prerequisite knowledge for Q38: Compound Angle formulae

Question 38. Prove that

a)

$$|z_1 z_2| = |z_1||z_2|$$

and

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2$$

b)

$$\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$$

Question 39. Prove that, given z is non-zero,

$$\arg\left(\frac{1}{z}\right) = -\arg z$$

Question 40. Given that

$$z = 3\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

find iz^2 writing your answer in principal modulus-argument form.

Question 41. It is given that $z = a + bi$ where a and b are real constants. It is additionally known that

$$|z^6| = 729 \text{ and } \arg(z^6) = \frac{5\pi}{3}$$

Determine the value of a and b giving your answer in exact form.

Question 42. On the Argand diagram, the points A and B are given by

$$z = 2\sqrt{3} + 2i \text{ and } w = -\sqrt{2} + \sqrt{2}i$$

respectively.
By writing z and w in modulus-argument form, find the geometrical transformations that map A onto B

Question 43. On the Argand diagram, the points A and B are given by

$$z = 1 + 4i \text{ and } w = -3 + 2i$$

respectively.
Find the multiplication scale factor and angle of rotation that maps A onto B

Question 44. On the Argand diagram, the points A and B are given by

$$z = 4 + 2i \text{ and } w = 1 - 5i$$

respectively. Find the multiplication scale factor and angle of rotation that maps A onto B

Question 45. For two complex numbers, z and w it is known that

$$\arg(z) = \frac{\pi}{6} \text{ and } \arg(w) = \frac{2\pi}{3}$$

Sketch and label the points representing z, w and zw on an Argand diagram for each of the following cases:

a) $|z| = 2$, $|w| = 3$

b) $|z| = 4$, $|w| = 1$

Question 46. For two complex numbers, z and w it is known that

$$|z| = 2 \text{ and } |w| = 1$$

Sketch and label the points representing z, w and zw on an Argand diagram for each of the following cases:

a)

$$\arg(z) = \frac{\pi}{4} \text{ and } \arg(w) = \frac{2\pi}{5}$$

b)

$$\arg(z) = \frac{\pi}{6} \text{ and } \arg(w) = -\frac{5\pi}{8}$$

Question 47. Given that

$$z_1 = 3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

and

$$z_2 = 3\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

a) sketch z_1 and z_2 on the same Argand diagram, labelling the points A and B respectively.

A third complex number z_3 which satisfies $|z_3| = 3$ and $-\pi < \arg(z_3) < 0$ and is plotted on the same Argand diagram as point C

b) Find the angle ACB

Question 48. It is known that z and w are complex numbers.

a) Show that $|z|^2 = zz^*$

b) Show that $(z - w)^* = z^* - w^*$

c) Find $|z - w|^2$ in terms of z, z^*, w and w^*

It is given that $|z + w| = 4$, $|z| = 2$ and $|w| = 3$

d) Find $|z - w|$

Question 49. It is known that z and w are complex numbers. Show that

$$|z+w|^2 + |z-w|^2 = 2|z|^2 + 2|w|^2$$

Question 50. Given

$$z = k(\cos\theta + i\sin\theta)$$

where k is real,

a) show that $|z| = k$

b) Find $\arg(z+k)$ in terms of θ

Question 51. The complex numbers z and w are such that

$$z = 2 + 2\sqrt{3}i$$

and

$$w = 1 + i$$

a) Find $\frac{z}{w}$ in the form $a + bi$
b) Write z in the form $r(\cos\theta + i\sin\theta)$
c) Write w in the form $r(\cos\theta + i\sin\theta)$
d) Find $\frac{z}{w}$ in the form $r(\cos\theta + i\sin\theta)$
e) Hence show that

$$\sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

Question 52. The complex numbers z and w are such that

$$z = 4\sqrt{3} - 4i$$

$$\arg(w) = \frac{\pi}{4} \text{ and } |w| = 4$$

a) Find w in the form $a + bi$
b) Hence find $\frac{z}{w}$ in the form $a + bi$
c) Write z in the form $r(\cos\theta + i\sin\theta)$
d) Find $\frac{z}{w}$ in the form $r(\cos\theta + i\sin\theta)$
e) Hence show that

$$\cos\left(-\frac{5\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$$

Question 53. For each of the following cases, find the smallest value of n such that $\arg((wz)^n) = 0$ and state $|(wz)^n|$ for that value of n:

a) $z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$, $w = 5\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

b) $z = 3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$, $w = 8\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

c) $z = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$,

$$w = 3\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right)$$

Question 54. Given that n is a positive integer and

$$\left(\frac{1+\sqrt{3}i}{2-2i}\right)^n$$

is a real number, find the smallest possible value of n